F. Computational Requirements

Method	Generator Compute	Classifier Compute	Overall Compute	Inference Throughput*	N_{params}	FID↓	IS↑	Precision↑	Recall↑
LSUN Churches 256^2									
StyleGAN2 [42]†	64	-	64	-	59M	3.86	-	-	-
LDM-8 (ours, 100 steps, 410K)	18	-	18	6.80	256M	4.02	-	0.64	0.52
LSUN Bedrooms 256^2									
ADM [15]† (1000 steps)	232	-	232	0.03	552M	1.9	-	0.66	0.51
LDM-4 (ours, 200 steps, 1.9M)	60	-	55	1.07	274M	2.95	-	0.66	0.48
CelebA-HQ 256^2									
LDM-4 (ours, 500 steps, 410K)	14.4	-	14.4	0.43	274M	5.11	-	0.72	0.49
FFHQ 256^2									
StyleGAN2 [42]	32.13‡	-	32.13†	-	59M	3.8	-	-	-
LDM-4 (ours, 200 steps, 635K)	26	-	26	1.07	274M	4.98	-	0.73	0.50
ImageNet 256^2									
VQGAN-f-4 (ours, first stage)	29	-	29	-	55M	0.58††	-	-	-
VQGAN-f-8 (ours, first stage)	66	-	66	-	68M	1.14††	-	-	-
BigGAN-deep [3]†	128-256		128-256	-	340M	6.95	$203.6_{\pm 2.6}$	0.87	0.28
ADM [15] (250 steps) †	916	-	916	0.12	554M	10.94	100.98	0.69	0.63
ADM-G [15] (25 steps) †	916	46	962	0.7	608M	5.58	-	0.81	0.49
ADM-G [15] (250 steps)†	916	46	962	0.07	608M	4.59	186.7	0.82	0.52
ADM-G,ADM-U [15] (250 steps)†	329	30	349	n/a	n/a	3.85	221.72	0.84	0.53
LDM-8-G (ours, 100, 2.9M)	79	12	91	1.93	506M	8.11	$190.4_{\pm 2.6}$	0.83	0.36
LDM-8 (ours, 200 ddim steps 2.9M, batch size 64)	79	-	79	1.9	395M	17.41	72.92	0.65	0.62
LDM-4 (ours, 250 ddim steps 178K, batch size 1200)	271	-	271	0.7	400M	10.56	$103.49_{\pm 1.24}$	0.71	0.62
LDM-4-G (ours, 250 ddim steps 178K, batch size 1200, classifier-free guidance [32] scale 1.25)	271	-	271	0.4	400M	3.95	$178.22_{\pm 2.43}$	0.81	0.55
LDM-4-G (ours, 250 ddim steps 178K, batch size 1200, classifier-free guidance [32] scale 1.5)	271	-	271	0.4	400M	3.60	$247.67_{\pm 5.59}$	0.87	0.48

Table 18. Comparing compute requirements during training and inference throughput with state-of-the-art generative models. Compute during training in V100-days, numbers of competing methods taken from [15] unless stated differently;*: Throughput measured in samples/sec on a single NVIDIA A100;†: Numbers taken from [15] ;‡: Assumed to be trained on 25M train examples; ††: R-FID vs. ImageNet validation set

In Tab 18 we provide a more detailed analysis on our used compute ressources and compare our best performing models on the CelebA-HQ, FFHQ, LSUN and ImageNet datasets with the recent state of the art models by using their provided numbers, *cf*. [15]. As they report their used compute in V100 days and we train all our models on a single NVIDIA A100 GPU, we convert the A100 days to V100 days by assuming a $\times 2.2$ speedup of A100 vs V100 [74][4]. To assess sample quality, we additionally report FID scores on the reported datasets. We closely reach the performance of state of the art methods as StyleGAN2 [42] and ADM [15] while significantly reducing the required compute resources.

[4]This factor corresponds to the speedup of the A100 over the V100 for a U-Net, as defined in Fig. 1 in [74]

G. Details on Autoencoder Models

We train all our autoencoder models in an adversarial manner following [23], such that a patch-based discriminator D_ψ is optimized to differentiate original images from reconstructions $\mathcal{D}(\mathcal{E}(x))$. To avoid arbitrarily scaled latent spaces, we regularize the latent z to be zero centered and obtain small variance by introducing an regularizing loss term L_{reg}.
We investigate two different regularization methods: (i) a low-weighted Kullback-Leibler-term between $q_\mathcal{E}(z|x) = \mathcal{N}(z; \mathcal{E}_\mu, \mathcal{E}_{\sigma^2})$ and a standard normal distribution $\mathcal{N}(z; 0, 1)$ as in a standard variational autoencoder [46, 69], and, (ii) regularizing the latent space with a vector quantization layer by learning a codebook of $|\mathcal{Z}|$ different exemplars [96].
To obtain high-fidelity reconstructions we only use a very small regularization for both scenarios, *i.e.* we either weight the $\mathbb{KL}$ term by a factor $\sim 10^{-6}$ or choose a high codebook dimensionality $|\mathcal{Z}|$.

The full objective to train the autoencoding model $(\mathcal{E}, \mathcal{D})$ reads:

$$L_{\text{Autoencoder}} = \min_{\mathcal{E},\mathcal{D}} \max_{\psi} \Big(L_{rec}(x, \mathcal{D}(\mathcal{E}(x))) - L_{adv}(\mathcal{D}(\mathcal{E}(x))) + \log D_\psi(x) + L_{reg}(x; \mathcal{E}, \mathcal{D}) \Big) \tag{25}$$

DM Training in Latent Space Note that for training diffusion models on the learned latent space, we again distinguish two cases when learning $p(z)$ or $p(z|y)$ (Sec. 4.3): (i) For a KL-regularized latent space, we sample $z = \mathcal{E}_\mu(x) + \mathcal{E}_\sigma(x) \cdot \varepsilon =: \mathcal{E}(x)$, where $\varepsilon \sim \mathcal{N}(0, 1)$. When rescaling the latent, we estimate the component-wise variance

$$\hat{\sigma}^2 = \frac{1}{bchw} \sum_{b,c,h,w} (z^{b,c,h,w} - \hat{\mu})^2$$

from the first batch in the data, where $\hat{\mu} = \frac{1}{bchw} \sum_{b,c,h,w} z^{b,c,h,w}$. The output of $\mathcal{E}$ is scaled such that the rescaled latent has unit standard deviation, *i.e.* $z \leftarrow \frac{z}{\hat{\sigma}} = \frac{\mathcal{E}(x)}{\hat{\sigma}}$. (ii) For a VQ-regularized latent space, we extract z *before* the quantization layer and absorb the quantization operation into the decoder, *i.e.* it can be interpreted as the first layer of $\mathcal{D}$.

H. Additional Qualitative Results

Finally, we provide additional qualitative results for our landscapes model (Fig. 12, 23, 24 and 25), our class-conditional ImageNet model (Fig. 26 - 27) and our unconditional models for the CelebA-HQ, FFHQ and LSUN datasets (Fig. 28 - 31). Similar as for the inpainting model in Sec. 4.5 we also fine-tuned the semantic landscapes model from Sec. 4.3.2 directly on 512^2 images and depict qualitative results in Fig. 12 and Fig. 23. For our those models trained on comparably small datasets, we additionally show nearest neighbors in VGG [79] feature space for samples from our models in Fig. 32 - 34.

Figure 19. *LDM-BSR* generalizes to arbitrary inputs and can be used as a general-purpose upsampler, upscaling samples from the LSUN-Cows dataset to 1024^2 resolution.